



GROUPED HYDRO POWER PROJECT BY EKI ENERGY SERVICES LIMITED



Document Prepared by EKI Energy Services Limited

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CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Sectoral Scope and Project Type.....	5
1.3	Project Eligibility.....	6
1.4	Project Design	8
1.5	Project Proponent	10
1.6	Other Entities Involved in the Project	11
1.7	Ownership.....	11
1.8	Project Start Date	12
1.9	Project Crediting Period	12
1.10	Project Scale and Estimated GHG Emission Reductions or Removals	12
1.11	Description of the Project Activity	13
1.12	Project Location.....	17
1.13	Conditions Prior to Project Initiation	18
1.14	Compliance with Laws, Statutes and Other Regulatory Frameworks	18
1.15	Participation under Other GHG Programs	19
1.16	Other Forms of Credit.....	20
1.17	Sustainable Development Contributions.....	20
1.18	Additional Information Relevant to the Project.....	24
2	SAFEGUARDS	24
2.1	No Net Harm	24
2.2	Local Stakeholder Consultation	24
2.3	Environmental Impact	26
2.4	Public Comments	27
2.5	AFOLU-Specific Safeguards	27
3	APPLICATION OF METHODOLOGY.....	27
3.1	Title and Reference of Methodology	27
3.2	Applicability of Methodology	28
3.3	Project Boundary	31
3.4	Baseline Scenario	32

3.5	Additionality	33
3.6	Methodology Deviations	47
4	IMPLEMENTATION STATUS	47
4.1	Implementation Status of the Project Activity	47
5	ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS.....	51
5.1	Baseline Emissions	51
5.2	Project Emissions	57
5.3	Leakage.....	58
5.4	Estimated Net GHG Emission Reductions and Removals.....	59
6	MONITORING	60
6.1	Data and Parameters Available at Validation.....	60
6.2	Data and Parameters Monitored.....	63
6.3	Monitoring Plan.....	64
7	QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	66
7.1	Data and Parameters Monitored.....	66
7.2	Baseline Emissions	66
7.3	Project Emissions	66
7.4	Leakage.....	67
7.5	Net GHG Emission Reductions and Removals	67

1 PROJECT DETAILS

1.1 Summary Description of the Project

The project serves the purpose of producing clean electrical energy in a sustainable manner. The project activity involves implementation of 2 turbines of one is 9.51 MW Mai Beni Hydropower Project (MBHP) and another is 6.2 MW Lower Jogmai Khola Hydropower Project (LJKHP) which utilize potential energy available for power generation. Therefore, no fossil fuels are involved for power generation in the project activity. The proposed project activity will also reduce the anthropogenic emissions of greenhouse gases into the atmosphere by avoiding business as usual operation of existing fossil fuels-based power plants to produce equivalent amount of electricity. This is a grouped project and the new instances will be added in the VCS project in the future in Nepal.

The details of the 9.51 MW Mai Beni Hydropower Project (MBHP) are mentioned below –

Mai Beni Hydropower Project is a run-of-river (ROR) project located in Ilam District, Province 1 of Nepal. The project will have installed capacity of 9.51MW.

Mai Beni Hydropower Project is located within Mai-Jogmai Rural Municipality and Ilam Municipality, Ilam District, State-1 in the Eastern Region of Nepal. The project site is about 527 km east of Kathmandu. The headworks site is located near the border of Soyang and Namsaling. The powerhouse is located 900m downstream of Mai khola and Jogmai khola confluence.

The electricity generated is supplied to Integrated Nepal Power System (INPS) through 4 km long, 33 kV Transmission line from the project to the 132/33kV Godak sub-station of NEA (Nepal Electricity Authority).

The details of the 6.2 MW Lower Jogmai Khola Hydropower Project (LJKHP) are mentioned below–

The project activity is a run of the river (ROR) scheme at Jogami River of Ilam District, Mechi Zone of Eastern Development Region, Nepal. Project site is accessible by Ilam-Namsaling motorable road from Tilkenimode to Powerhouse Site which is about 6 km. The same road is reaching to surge tank which is about 7.25 km from Tilkenimode. The powerhouse can also be reached by motorable road about 3.5 km from Silpur which is opposite bank of the powerhouse. Project has constructed access road from powerhouse to headworks.

The electricity generated is supplied to Integrated Nepal Power System (INPS) through 4 km long, 33 kV Single Circuit Transmission line from the project to the 132/33kV Godak sub-station of NEA.

The location of the project

The project location for Project Activity Instances is in Nepal and details are provided in Section 1.12 of this document. The project location details, KML files for Nepal country are submitted along with Joint PDMR.

A brief description of the scenario existing prior to the implementation of the project

The baseline scenario would be either continued use of fossil fuel dominated grid connected electricity generation or fossil fuel used in power plant or carbon intensive electricity.

An estimate of annual average and total GHG emission reductions and removals

During registration of grouped project activity, both two instances are included. The average annual GHG emission reduction from the first project activity instance is 21,741 tCO₂e.

The current monitoring period was from 17-September-2021 to 31-December-2022 (first and last days included) and the total emission reduction is 26,567 tCO₂e.

Project Activity instance:

The details of the first/current project activity instance are as below-

S.I No.	Customer	Installed Capacity (MW)	Date of Commissioning
1	Samling Power Company Ltd.	9.51	17-September-2021
2	Asian Hydropower Ltd.	6.20	01-November-2021
	Total	15.71	

<u>Audit Type</u>	<u>Period</u>	<u>Program</u>	<u>VVB Name</u>	<u>Number of years</u>
Validation	<u>17-September-2021 to 16-September-2028</u>	<u>VCS</u>	<u>TUV SUD</u>	7 years
Verification	<u>17-September-2021 to 31-December-2022</u>	<u>VCS</u>	<u>TUV SUD</u>	1 year, 3 months, 14 days
<u>Total</u>				Validation – 7 years Verification - 1 year, 3 months, 14 days

1.2 Sectoral Scope and Project Type

The project activity falls under consideration of the grouped project activity. The project activity instances as part of the grouped project will have the following parameters.

Sectoral Scope : 01 - Energy industries (renewable / non-renewable sources)
 Project Type : I - Renewable Energy Projects
 Project Category : Each project activity instances will apply only small-scale hydro project
 Methodology : AMS-I.D.: Grid connected renewable electricity generation – Version 18.0

The project activity under consideration is a grouped project activity.

1.3 Project Eligibility

The project involves the electricity generation from the hydro power plant which falls under the Grid-connected electricity generation activities using hydroelectric power plants (eligible under small scale for LDC's). Therefore, it is eligible under the scope of the VCS Program. As per VCS Standard Version 4.5¹, section 2.1.3.

The following table demonstrates how the initial project activity instances being included in the grouped project activity fulfill above mentioned eligibility conditions:

Sr. No	Eligibility Criteria	Project Activity instances eligibility
1	Applicability Conditions: The project activity instances shall meet applicability conditions for applicable methodology (AMS I.D version 18.0) as defined in section 3.2.	All project activity instances meet applicability conditions of methodology (AMS I.D Version 18.0 (project activity instances having capacity less than 15 MW), hence this eligibility criteria are fulfilled. Please refer section 3.2 of this document for methodology applicability criteria.
2	Geographical Area: The project activity instances to be included in the grouped project activity will be activities involving Renewable Electricity Generation and grid using hydropower plant located within Nepal.	All project activity instances being included in the grouped project are located within geographic boundaries of Nepal. Hence this condition is fulfilled. Please refer commissioning certificate for Geographical area of project activity.
3	Baseline scenario: All Project Activity Instances shall meet the baseline definition as defined in respective methodology and as explained in section 3.4.	All initial project activity instances have same baseline as per methodology as detailed in subsequent sections. Hence this eligibility criterion is fulfilled. Please refer section 3.4 of this document for baseline scenario.

¹ <https://verra.org/wp-content/uploads/2023/08/VCS-Standard-v4.5-updated-11-Dec-2023.pdf>

4	Technology type: The project activity instances to be included in the grouped project activity will be Greenfield activities involving Renewable Electricity Generation and grid using hydropower. The electricity should be supplied to grid.	All project activity instances are hydro power plants supplying electricity for grid. All are green field projects. Hence this condition is fulfilled.
5	Additionality: The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in respective methodologies - for Small Scale Project instances the methodology applicable will be AMS-I.D.	The additionality for the all-project activity instances included in the grouped project has been demonstrated in section 3.5 of Joint PDMR.
6	Start Date: The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence.	The start date of grouped project is considered as the earliest commissioning date of project activity instance and start date of all other project activity instances is after this date. Please refer commissioning certificate for start date of grouped project activity.
7	Conditions that avoid double counting of emission reductions like unique identifications of project and claiming emission reduction only under one GHG program.	The existing and new project activity instances are identified by latitude and longitude and location of the plant. Declaration has been provided and these project activity instances are not participating in any other GHG program. Hence this condition is fulfilled.
8	The Grouped Project specific requirements stipulated by the Entity responsible for coordinating and managing grouped project for conducting local stakeholder consultations and environmental impact assessment (EIA), as applicable.	Local stakeholder consultation has been conducted as a part of grouped project activity. There is requirement for carrying out IEE for these projects in Nepal. According to the environmental report ² hydropower projects can conduct either IEE or EIA and details are provided in section 2.3; hence this condition is fulfilled.
9	The project activity using hydropower project will supply electricity to grid.	The existing and new project activity instances are hydro power plants supplying electricity to grid. Hence this condition is fulfilled.

² <https://www.sciencedirect.com/science/article/pii/S2667010021002857>

As per clause 3.6.8 of VCS Standard Version 4.5, The project proponent shall include in a singular project all project activity instances within ten kilometers of another instance of the same project activity and with the same project proponent (i.e., instances of the same project activity may not be spread across more than one project if they are within ten kilometers of each other). Thus, the distance between the two instances is more than 15 km, hence this requirement is fulfilled.

As per clause 3.6.12 of VCS Standard Version 4.5, “As with non-grouped projects, grouped projects may incorporate multiple project activities (see Section 3.6.1 – 3.6.3 for more information on multiple project activities). Where a grouped project includes multiple project activities, the project description shall designate which project activities may occur in each geographic area”.

As per para 3.6.9 of VCS Standard V4.5, each project activity instances are not exceeding one percent of the capacity limit shall be identified. The project activity instance involves multiple project activity instances and it is not cluster of project activity instances. Thus, there is no any requirement of division of clusters of project activity instances and no requirement to demonstrate that each project activity instances are not within one kilometer of each other.

1.4 Project Design

- ☐ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☒ The project is a grouped project

Eligibility Criteria

The project is a grouped project and there is no additional information needed to be provided. Following are the eligibility criteria for inclusion of new project activity instances into the grouped project activity as per para 3.6.16 of VCS Project standard v4.5:

Sr. No	Eligibility Criteria	Project Activity instances eligibility
1	Meet the applicability conditions set out in the methodology applied to the project.	All project activity instances meet applicability conditions of methodology (AMS I.D Version 18.0 (project activity instances having capacity less than 15 MW), hence this eligibility criteria are fulfilled. Please refer section 3.2 of this document for methodology applicability criteria.

2	Use the technologies or measures specified in the project description.	All project activity instances are hydro power plants supplying electricity for grid. All are green field projects. Hence this condition is fulfilled.
3	Apply the technologies or measures in the same manner as specified in the project description.	All project activity instances are hydro power project. Hence it is applicable.
4	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	All initial project activity instances have same baseline as per methodology as detailed in subsequent sections. Hence this eligibility criterion is fulfilled. Please refer section 3.4 of this document for baseline scenario.
5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area. For example, the new project activity instances have financial, technical and/or other parameters (such as the size/scale of the instances) consistent with the initial instances, or face the same investment, technological and/or other barriers as the initial instances.	All project activity instances are hydro power plants. All are green field projects. Located in the same geographical area (Nepal) also only the small-scale projects are considered in the project activity with similar financial and technical parameters. Hence this eligibility criterion is fulfilled.

Along with the above points, the Inclusion of New Project Activity Instance follows the below criteria of Grouped projects provide for the inclusion of new project activity instances subsequent to the initial validation of the project.

Sr. No	Eligibility Criteria	Project Activity instances eligibility
1	Occur within one of the designated geographic areas specified in the project description.	All project activity instances meet applicability conditions of methodology (AMS I.D Version 18.0 (project activity instances having capacity less than 15 MW), hence this eligibility criteria are fulfilled. Please refer section 3.2 of this document for methodology applicability criteria.
2	Conform with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial conformance with multiple sets of eligibility criteria is insufficient.	New project activity instances will comply with a complete set of eligibility criteria as mentioned.

3	Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate conformance with the applicable set of eligibility criteria and enable evidence gathering by the validation/verification body.	New project activity instances information to be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria.
4	Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).	New project activity instances ownership needs to be evidenced from the start date of the respective project activity instance.
5	Have a start date that is the same as or later than the grouped project start date.	New project activity instances' start date should be after the start date of grouped project activity.
6	Only be eligible for crediting from the later of start date of the project activity instance or the start of the verification period in which they were added to the grouped project, through to the end of the total project crediting period.	New project activity instances are eligible to claim credits from the start date of the project activity instance through to the end of the project crediting period.
7	Not be or have been enrolled in another VCS project.	New project activity instances will not be previously registered or currently enrolled in any another VCS projects.
8	Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 – 3.6.9.	Each project activity instances are less than 15 MW. Capacity of instances are 9.51 MW & 6.2 MW, which is less than 15 MW.

1.5 Project Proponent

Organization name	Urja Developers Private Limited
Contact person	Bhanu Bhakta Pokharel
Title	Director
Address	Krishna Galli-3, Pulchowk, Lalitpur, Nepal
Telephone	977-9851059177
Email	bhanu.pokharel@urjadeveloper.com

Organization name	EKI Energy Services Limited
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Contact person	Manish Dabkara
Title	Managing Director & Chief Executive Officer
Address	Office No 201, Plot No 48, Scheme 78, Vijay Nagar Part- II, Indore 452010, India
Telephone	+91 99075 34900
Email	manish@enkingint.org and registry@enkingint.org

1.6 Other Entities Involved in the Project

Organization name	EKI Energy Services Limited
Role in the project	Project Consultant
Contact person	Mr. Ankit Sethiya
Title	Deputy Manager – Operations
Address	Plot No 48, Scheme 78, Vijay Nagar Part- II, Indore 452010, Madhya Pradesh, India
Telephone	0731 428 9086
Email	ankit.sethiya@enkingint.org

1.7 Ownership

The renewable energy technology installed at site of each of the project activities as part of the proposed project activity are under legal ownership of Samling Power Company Ltd. & Asian Hydropower Ltd. EKI Energy Services Limited & Urja Developers Private Limited are authorized PP by all owners of hydro power plant. All the carbon credits related rights are with EKIESL; and EKIESL will be the concerned authority for all modalities and procedure with respect to VCU of the project activity. The ownership of VCU will be with EKIESL. Legal Ownership can be verified with commissioning certificate. Further as per agreement between Urja Developers Private Limited and EKI the VCU rights are transferred. Samling Power Company Ltd. & Asian Hydropower Ltd. has authorized Urja Developers Private Limited and Urja Developers Private Limited has transferred all rights of carbon credit to EKIESL.

This demonstrates the right of use according to clause 3.7.1 (3) of VCS Standard (v4.5) – “a project ownership arising by virtue of a statutory, property or contractual right in the plant,

equipment or process that generates GHG emission reductions and/or removals (where the project proponent has not been divested of such project ownership)”.

1.8 Project Start Date

Start date of project activity is 17-September-2021, which is the date of commissioning of the earliest commissioned project involved in the grouped project activity. PP is willing to include the project activity instances during registration having commissioning dates from 17-September-2021 onward and as per deadline of 2 years of validation completion from date of commissioning.

1.9 Project Crediting Period

Crediting Period Start date: 17-September-2021

Crediting Period End date: 16-September-2028

The grouped project activity adopts renewable crediting period of 7 years period which can be renewed for maximum 2 times. This start date of crediting period will vary based on actual earliest commissioned date within the project activity instances of grouped project activity. PP declares that next instance crediting period start date for crediting period is already established. Hence the guideline for the new instance section 3.6.17 of VCS standard v 4.5 will be followed.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☒ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☐ >1,000,000 tCO₂e/year

Estimated average annual GHG emission reductions from the project activity for first crediting period are 21,741 tCO₂e per year, which is less than 300,000 tCO₂e per year as per the section no 3.10.1 of VCS standard version 4.5. Hence the project scale of the project activity will become “Projects” as shown below:

Project Scale

Project	✓
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1	21,741
Year 2	21,741
Year 3	21,741
Year 4	21,741
Year 5	21,741
Year 6	21,741
Year 7	21,741
Total estimated ERs	152,187
Total number of crediting years	07
Average annual ERs	21,741

1.11 Description of the Project Activity

The project involves installation of Mai Beni Hydropower Project and Lower Jogmai Khola Hydropower Project.

1. Mai Beni Hydropower Project is a run-of-river (ROR) project located in Illam District, Province 1 of Nepal. The project will have installed capacity of 9.51MW. A power purchase agreement has also been signed between Samling Power Company and Nepal Electricity Authority, the Government of Nepal undertaking company. The intake site is located near the border of Soyang and Namsaling. The powerhouse is located just downstream of Mai khola and Jogmai khola confluence. Overall project layout is proposed within a stretch of 5-6 km with all components located at the right bank of Mai Khola.

2. Lower Jogmai Khola Hydropower Project is a Run-of-River project, it is essential to conduct long-term hydrological investigation as well as to estimate the low flow and flood flow. As mentioned earlier, Jogmai is an ungauged River. Therefore, data from other River gauge stations has been taken to derive the long-term flow of Jogmai Khola at the intake site.

Installed capacity of the project is optimized at 6.2 MW. This scheme comprises of 20m wide diversion weir with weir crest level at 649 masl, which will divert the design flow into the intake located at the left side of the river. Weir will be built with a layer of boulder lining. An Under sluice having 2m width has been provided before the intake. Sediment settling system of the plant is located at the distance of about 200m from the intake. It has two chambers of size 62m x 6.05m x 6m (L x B x H). The scheme comprises of a 2870m long headrace pipe having diameter of 1.8m. A 55m long surge pipe of 3.0m diameter is designed to arrest the surge in the scheme. Powerhouse is connected by 417m long penstock having 1.8m diameter. Two Horizontal-axis Francis turbines each of 3.10 MW capacity will be installed in the powerhouse.

As per para 3.6 of VCS Standard V.4.5 (Project activity and project activity instance both have the specific meanings that are set out in the VCS Program document Program Definitions), each project activity instance of grouped project activity is less than 15 MW capacity. Thus, small scale methodology AMS I.D is applicable for proposed grouped project activity.

As per the applicable methodology, a Greenfield power plant is defined as “a new renewable energy power plant that is constructed and operated at a site where no renewable energy power plant was operated prior to the implementation of the project activity”.

The Project activity is a new facility (Greenfield) and the electricity generated by the Project will be exported to the national Grid. The Project will therefore displace an equivalent amount of electricity which would have otherwise been generated by fossil fuel dominant electricity grid. The Project Proponent plans to avail the VCS benefits for the Project.

Technical Specification for Sampling Power Company Ltd.

1. Turbine
 - Type: Horizontal shaft Francis turbine
 - Make: TPSC-I
 - Number: 2
 - Rated output capacity per unit: 4930 KW
 - Net Head: 136.64 m
 - Efficiency (max): 92.8% at minimum head
 - Turbine axis level: 495.58 masl
 - Rated Speed: 750 RPM
 - Runaway speed: 1380 RPM
2. Main Inlet Valve:
 - Make: Fouress Engineering (India) Ltd.
 - Type: Butterfly valve
 - Size: 900 mm
 - Design Pressure: 19.83 Kg/cm²

- Maximum flow: 4.471 m³/sec
- Method of Closing: Counter weight
- Method of opening: Oil pressure
- Valve Operating time:
 - 1) Normal closing time: 45-60 (adjustable)
 - 2) Normal opening time: 45-60 (adjustable)
- 3. Generator:
 - Type: Cylindrical Pole, Synchronous Generator
 - Make: TDPS
 - Rated Output capacity per unit: 5597 KVA + 10% COL
 - Power factor: 0.85
 - Voltage: 6.6 KV
 - Frequency: 50 Hz
 - No of units: 2
 - Excitation system: Brushless excitation
 - Efficiency: 96.8 % at 100% of full load 0.85 PF
 - Speed: 750 RPM
- 4. Oil Pressure Unit (OPU):
 - Make: Hydac (India) Pvt. Ltd.
 - Working Pressure: 120 Bar.
 - Test Pressure: 180 Bar
 - No of Pump: 2 (One working and another in Standby)
 - Rating of Pump: 2.2 KW, 400 V, 50 Hz, 1500 RPM
 - Main Pump Flow: 5 lpm
 - Piping material: Carbon Steel
 - Oil reservoir Capacity: 250 Liters
 - Working Fluid: Mineral Oil ISO VG 46
 - Viscosity of Oil @ 40 Deg. C: 46 cst
 - Operating maximum temperature: 80 deg
- 5. Lube oil system (LOS):
 - Make: Pascal System technik Pvt. Ltd.
 - No of pump: 2 AC & 1 DC
 - Rating of AC Pump: 3 HP, 400 V AC, 1445 RPM
 - Rating of DC Pump: 3 HP, 110 V DC, 1500 RPM
 - NDE flow rate: 7 LPM
 - DE Flow rate: 57 LPM
 - Oil Tank Capacity: 630 Liters
- 6. DC System:

Battery Charger

 - Make: Universal Instruments Mfg. Co. Pvt. Ltd.
 - Capacity: 110 V, 250 Ah
 - No of Charger: 2 (1 main & 1 standby) FCBC
 - Float Voltage per cell: 2.23 +/- 0.01 Volt at 25-34 Deg. C temperature
 - Battery Type: VRLA Battery
 - Battery make: EXIDE Industries Ltd.
 - Rating of battery: 54 * 2 V / 250Ah

Technical Specification for Asian Hydropower Ltd.**1. Turbine**

- Type: Horizontal shaft Francis turbine
- Make: Flovel Energy Private Limited, India
- Number: 2
- Rated output capacity per unit: 3213 KW
- Net Head: 125.49 m
- Rated Discharge: 2.86 m³/s per unit
- Efficiency (max): 93.2%
- Turbine axis level: 514.24 masl
- Rated Speed: 750 RPM
- Runaway speed: 1215 RPM

2. Main Inlet Valve:

- Make: TB Hydro Flovel Valves Private Limited, India.
- Type: Butterfly valve
- Size: 900 mm
- Design Pressure: 18 bar
- Method of Closing: Counter weight
- Method of opening: Oil pressure
- Number of seals: 1 No.
- Valve Operating time:

Normal closing time: 45-60 (adjustable)

Normal opening time: 45-60 (adjustable)

3. Generator:

- Type: Cylindrical Pole, Synchronous Generator
- Make: WEG Industries Pvt. Ltd., India
- Rated Output capacity per unit: 3647 kVA + 10% COL
- Power factor: 0.85
- Voltage: 6.6 KV
- Frequency: 50 Hz
- No of units: 2
- Excitation system: Brushless excitation
- Efficiency: 96.6 % at 100% of full load 0.85 PF
- Speed: 750 RPM

4. Oil Pressure Unit (OPU):

- Make: Hydac (India) Pvt. Ltd.
- Working Pressure: 120 Bar.
- Test Pressure: 180 Bar
- No of Pump: 2 (One working and another in Standby)
- Rating of Pump: 3.7 kW, 400 V, 50 Hz, 1440 RPM
- Main Pump Flow: 8 lpm
- Piping material: Carbon Steel
- Oil reservoir Capacity: 200 Liters
- Working Fluid: Mineral Oil ISO VG 46/68
- Operating maximum temperature: 80 deg

5. Lube oil system (LOS):

- Make: Dropco Multilub Systems Pvt. Ltd.
- No of pump: 2 AC & 1 DC
- Rating of AC Pump: 1.1 kW, 400 V AC, 1450 RPM
- Rating of DC Pump: 1.1 kW, 110 V DC, 1450 RPM
- NDE flow rate: 4 LPM
- DE Flow rate: 28 LPM
- Oil Tank Capacity: 400 Liters

6. DC System:

Battery Charger

- Make: Green Secure Energy Systems
- Capacity: 110 V, 250 Ah
- No of Charger: 2 (1 main & 1 standby) FCBC, SMPS based technology
- Float Voltage per cell: 2.25 Volt/cell
- Boost Voltage per cell: 2.3 Volt/cell
- Battery Type: VRLA Battery
- Battery make: HBL Power Systems Ltd.
- Rating of battery: 55 * 2 V / 250Ah

1.12 Project Location

The details of the project locations are mentioned in the table below:

S.I No.	Customer	Premise address where the unit is installed	Installed Capacity (MW)	Latitude	Longitude
1	Samling Power Company Ltd.	Ilam District Province 1, Nepal	9.51	26°55'35.75"N	87°56'52.82"E
2	Asian Hydropower Ltd.	Ilam District Province 1, Nepal	6.20	26°53'48.37"N	87°59'31.58"E

This is a grouped project activity and the Nepal is the project location. The KML file shared with the VVB.



A KML file have been submitted in order to indicate the project location and geographic boundaries.

1.13 Conditions Prior to Project Initiation

The project is a Greenfield hydro power project and does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction. Prior to the initiation of the project activity, the equivalent amount of electricity would have been drawn from grid connected or new power plants, in national Grid. The grid is predominantly coal based and therefore is a major source of carbon dioxide emissions in India³ and same is explained in the baseline scenario in accordance of the methodology. The main emission in the pre project scenario is the power plants connected to the national Grid, and main GHG involved is CO₂.

As per methodology, the baseline scenario would be either continued use of fossil fuel dominated grid connected electricity generation or fossil fuel used in power plant or carbon intensive electricity. Nepal is importing power from India, so Indian grid is mentioned.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

³ <https://powermin.gov.in/en/content/interconnection-neighbouring-countries>

The Project has received necessary approvals for development and commissioning for each plant from the state Nodal agencies and is in compliance to the local laws and regulations. Project is complying following laws. List of Project Related Policies, Acts and Regulations are as follows -

Legislations	Functions
Company Act 2053 (1996) ⁴	Registration and operation
Hydropower Development Policy 2058 (2001) ⁵	Guidelines for hydropower development
National Biodiversity Strategy and Action Plan 2071 B.S. (2014) ⁶	Management of Diversity
Electricity Act 2049 (1992) ⁷	Regulations for survey, generation, transmission and distribution of Small Hydropower power
Water Resources Act 2049 (1992) ⁸	Regulations for rational utilization, conservation,
NEA's policy regarding hydropower 2055 (1998) ⁹	Regulation regarding buy back rates and Power purchase agreement of small hydropower projects by private sectors
Environment Protection Act 2054 (1997) ¹⁰ Environment Protection Regulations (as amended) 2055(1999)	Regulations regarding environment protection and hydropower development
National Energy Crises Policy, 2073 ¹¹	To improve Energy Crisis in Nepal

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The proposed grouped project activity has not been registered and is not seeking registration atmoment under any other GHG programs.

The initial project activity instances are not participating in any other GHG program.

⁴ https://nepalindata.com/media/resources/bulkuploaded/companies_act_2053_eng_april_17.pdf

⁵ <https://www.lawcommission.gov.np/en/wp-content/uploads/2018/09/hydropower-development-policy-2058-2001.pdf>

⁶ <https://www.cbd.int/doc/world/np/np-nbsap-v2-en.pdf>

⁷ <https://admin.theiguides.org/Media/Documents/ElectricityAct1992.pdf>

⁸ <https://admin.theiguides.org/Media/Documents/WaterResourcesAct1992.pdf>

⁹ https://www.nea.org.np/admin/assets/uploads/supportive_docs/1998-99-compressed.pdf

¹⁰ <https://faolex.fao.org/docs/pdf/nep100343.pdf>

¹¹ [https://aepc.gov.np/uploads/docs/2018-06-19_RE%20Subsidy%20Policy,%202073%20\(English\).pdf](https://aepc.gov.np/uploads/docs/2018-06-19_RE%20Subsidy%20Policy,%202073%20(English).pdf)

1.15.2 Projects Rejected by Other GHG Programs

The proposed grouped project activity is not participating in other Environmental credits, other GHG programs and has not been rejected by other GHG programs.

The initial project activity instances are not participating in other Environmental credits, other GHG programs and have not been rejected by other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

The grouped project activity under consideration is not participating in any other Emission trading program and other binding limits.

The all-project activity instances are not participating at moment in any other emission trading program and other binding limits. The GHG emission reductions generated during the current monitoring period have not been used for compliance of other mechanism.

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes

☒ No

The proposed grouped project activity neither has nor intends to generate any other form of GHG related environmental credit for GHG emission reductions or removals claimed under the VCS Program. The initial project activity instances are neither has nor intends to generate any other form of GHG related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

1.16.3 Supply Chain (Scope 3) Emissions

Not Applicable

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

Contribution to sustainable development:

Apart from generation of renewable electricity, the grouped project activity and project activity instances proposed to be included under it would contribute to the sustainable development of the region - socially, environmentally and economically. Ministry of Environment and Forests, has stipulated economic, social, environment and technological well-being as the four indicators of sustainable development. The project contributes to sustainable development using the following ways.

Social well-being: The project would help in generating employment opportunities during the construction and operation phases. The project activity will lead to development in infrastructure in the region like development of roads and also may promote business with improved power generation.

Economic well-being: The project is a clean technology investment in the region, which would not have been taken place in the absence of the VCS benefits the project activity will also help to reduce the demand supply gap in the state. The project activity creates local employment generation which helps economic well-being of local people.

Technological well-being: The successful operation of project activity would lead to promotion of hydro based power generation and would encourage other entrepreneurs to participate in similar projects.

Environmental well-being: Hydro being a renewable source of energy, it reduces the dependence on fossil fuels and conserves natural resources which are on the verge of depletion. Due to its zero emission the Project activity also helps in avoiding significant amount of GHG emissions.

1.17.2 Sustainable Development Contributions Activity Monitoring

Designated National Authority (DNA) of the host country (Nepal) has stipulated two indicators for sustainable development in the interim approval guidelines for Clean Development Mechanism (CDM) projects in Nepal. The project proponent believes that the project activity has contributed to sustainable development in terms of the two indicators.

The project supplies clean electricity from the hydro power project to the national Grid, hence displacing the electricity generated from grid connected fossil fuel power plants and thereby avoiding the equivalent Carbon dioxide which is a Green House Gas.

The project activity fulfils SDG Target 7.2 By 2030, increase substantially the share of renewable energy in the global energy mix,

8.6 By 2020, substantially reduce the proportion of youth not in employment, education or training and

13.2 Integrate climate change measures into national policies, strategies and planning of SDG Goals.

- SDG 7.0: According to SDG target 7.0, indicator 7.2.1, the project activity generates electricity from hydro energy sources that ensures supply of electricity sources from clean energy sources at affordable rate. In the 1st monitoring period, this project activity has supplied 91,678.73 MWh clean electricity.
- SDG 8.0: According to SDG target 8.0, indicator 8.6.1, the project activity creates together with the technology supplier, the project organize training for the staff on the technology and the monitoring of the plant operation, and the emergency and safety procedures. In this monitoring period, this project activity has achieved 02 trainings.
- SDG 13.0: According to SDG target 13.0, indicator 13.2.2, it states “Take urgent action to combat climate change and its impact”. By mitigating 26,567 tCO₂e which is greenhouse gas, the project activity has contributed towards SDG target 13.0. for present monitoring period.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7.2.1: Renewable energy share in the total final energy consumption	Implemented activities to increase	91,678.73 MWh of renewable electricity has been supplied to national grid during the reported period that helps to increase the renewable energy share in the energy mix.	About 525,115 MWh renewable electricity will be supplied to Nepal grid that helps to increase the renewable energy share in the energy mix.
2)	13.0	13.2.2: Total greenhouse gas emissions per year	Implemented activities to increase	By supplying 91,678.73 MWh clean electricity to national grid, the project avoided release of 26,567 tCO ₂ e in to the atmosphere during the reporting period.	Prevented the release of 152,187 tCO ₂ e into the atmosphere.
3)	8.0	8.6.1: Proportion of youth (aged 15-24 years) not in education, employment or training	Implemented activities to increase	The number of trainings provided are 02 numbers.	In total 14 number of trainings will be conducted during the project activity.

1.18 Additional Information Relevant to the Project

Leakage Management

Since the project activity instances to be included into the grouped project activity are renewable energy projects based on hydropower, no leakage emissions are considered - in line with paragraph 27 of the small-scale methodology AMS.I.D (version 18.0).

Thus, leakage management plan and implementation of leakage and risk mitigation measures are not applicable for this grouped project activity.

Commercially Sensitive Information

Not applicable. No any commercially sensitive information has been excluded from the public version of the project description. There is no commercially sensitive information.

Further Information

Not Applicable

2 SAFEGUARDS

2.1 No Net Harm

The project activity promotes environmental and socio-economic well-being as it results in zero GHG emissions due to installation and operation of clean, renewable energy technology for electricity generation.

Thus, there are no any significant impacts due to implementation of project activity on air, water, soil quality and ambience are envisaged due to the project activity.

2.2 Local Stakeholder Consultation

Samling Power Company Ltd.

The Local Stakeholder Meeting was organized for local stakeholder consultation at Ward no. 10 of Ilam Municipality of Ilam District on 17-March-2019. The stakeholders were informed regarding the local stakeholder meeting through public notice dated 12-March-2019. The following are the local stakeholders for the project activity:

- Local community
- Local village administration
- Students

- Shopkeepers
- Local vendors

In the introductory speech, the representatives of Project Owner welcomed the gathering and given a brief about the hydropower project activity. Subsequent to the introductory speech, stakeholders were explained about the electricity generation from hydro project is an environmental friendly power generation technology contributing to reduction in GHG emissions. They were also explained about the benefits of the hydro power projects like, increasing energy availability and improving quality of power and its assistance to the local population by providing employment opportunities to both skilled & unskilled labours. As a part of ongoing communication with local stakeholders, local peoples are informed about the grievance register placed at site. If any concerns are received during the operation of the project activity, the same will be addressed if relevant to the project activity. For the current monitoring period no grievances have been received.

The Minutes of LSH meeting along with List of Attendees and copy of public notice shall be submitted to the VVB. Feedback received from stakeholders-

1. During the implementation of the project, employment opportunities should be provided to skilled and unskilled workers at the local level.

Response - With the increased possibilities for employment of local youth due to the project activity, it was pointed out that preference would be given for locals in the employment opportunities.

2. If there will be any accident due to the project activity e.g. electric shock etc.

Response – There is no chance of such accident; any remote chance of such accident shall be identified and nullified.

3. Will there be free supply of power to the local people?

Response – Project promoter can't supply directly power to the local people. They have to get authorized connection from Govt. body. But due to the project activity the supply of power in the area will increase.

4. Will it impact the climatic condition of the nearby area?

Response – No, it will not impact the climatic condition of the nearby area.

Asian Hydropower Ltd.

The Local Stakeholder Meeting was organized for local stakeholder consultation at Ward no. 10 of Ilam Municipality of Ilam District on 06-February-2019. The stakeholders were informed regarding the local stakeholder meeting through public notice dated 29-January-2019. The following are the local stakeholders for the project activity:

- Local community
- Local village administration
- Students
- Shopkeepers
- Local vendors

In the introductory speech, the representatives of Project Owner welcomed the gathering and given a brief about the hydropower project activity. Subsequent to the introductory speech, stakeholders were explained about the electricity generation from hydro project is an environmental friendly power generation technology contributing to reduction in GHG emissions. They were also explained about the benefits of the hydro power projects like, increasing energy availability and improving quality of power and its assistance to the local population by providing employment opportunities to both skilled & unskilled labours.

The Minutes of LSH meeting along with List of Attendees and copy of public notice shall be submitted to the VVB. Feedback received from stakeholders-

1. The project will have to build a road.

Response – Yes, for better connectivity the roads will be repaired.

2. Locals should be given employment

Response – As per the requirements of the plant, preference would be given for locals in the employment opportunities.

2.3 Environmental Impact

For hydropower projects in Nepal the EIA (IEE) study has to comply with a range legal requirements of government of Nepal like Constitution of Nepal, 2072, Policy, Guidelines, Acts, Regulations and International Treaties and Conventions such as: Three Years man 2070 (2013), Hydropower Development Policy, 2001, Forest Policy, 2000, Nepal Biodiversity strategy, 2002, Nepal Biodiversity Strategy Implementation man, 2006, National Water strategy (2002) and Nepal Water man 2005, National Environmental Impact Assessment Guidelines, 1993, Department of Electricity Development Manuals, Forestry Sector ETA Guidelines 1995. Forest Produce, Collection, Sale and Distribution Guideline, 2000, Community Forest Guidelines, 2008, Community Forest Inventory Guidelines, 2005, Environmental Management Guidelines (Road), 1997, Guidelines for the physical Infrastructure Development and Operation in the Protected Areas 2065 (2008), soil and Watershed Conservation Act. 1982, Forest Act, Convention on Biological Diversity, 1992 etc.

The environmental impacts of this project are not considered to be significant. Nevertheless, a detailed Environmental Management Plan, which includes relevant mitigation measures, monitoring procedures and responsibilities, is provided in the Environmental Assessment Report.

- Topography, hydrological, geological and power optimization studies have shown that the proposed dam site and tailrace of this Hydropower Project are the most suitable ones. Hence, the project seems feasible from all perspectives.
- It is seen that wind, solar and bio-gas are not proper alternatives to the Hydro Power Plant.
- The project won't encroach into any the declared protected areas of Nepal, nor will it undermine the rich cultural heritage sites of Nepal.

Following impacts are identified as per IEE report submitted to Ministry of Energy, Water Resources and Irrigation through Department of Electricity Development (DoED) in the month of March 2019

with Letter no. 075/076. The analysis of those impacts is done and mitigation measures are identified as per IEE report and same is included in EMP.

1. Physical Impact – The project activities like excavation and slope cutting in project components like powerhouse, penstock pipe, and surge pipe will lead to change in land topography. Impact on the topography will also occur by the clearance of land.

Mitigation Measures – Change in land use due to construction of structural components of the project like headworks, access road and switchyard permanent camp for operation workers is unavoidable and remains as residual impact till the project exists. Only the changes due to project support facilities can be mitigated.

Land area required for the other support project features such as labour camp, storage yard will be taken on lease.

2. Biological Impacts – Impact on surrounding forest resources for its products like firewood, timber and NTFPs.

Mitigation Measures – In order to minimize the need of forest products, the project workers will be provided kerosene to meet their fuel requirements for cooking purpose. However, emphasize will be given to provide LPG to the construction workers for cooking purpose.

The project management and contractor will impose complete prohibition on the purchase, sale and storage of forest products. Anyone found with local forest products in the camps will be penalized. This is a management task and no extra cost is envisaged.

2.4 Public Comments

The project has been listed for 30 days period. This project was open for public comment from 24-March-2023 to 23-April-2023. No public comments were received during the public comment period.

2.5 AFOLU-Specific Safeguards

This section is not required as this project activity is a non-AFOLU project activity.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The project activity instances as part of the proposed grouped project activity will be small scale project activity instances with power generation capacity less than or equal to 15 MW as per CDM terminology. Accordingly, below mentioned methodology will be used for each of the project activity instances.

The project activity instance based on its capacity:

The methodology for the project activity instances will be the approved methodology for small-scale CDM project activities. The details of the methodology are as follows:

Methodology: AMS.I.D

Project Type : Type-I: Renewable Energy Projects
 Title : AMS I. D Grid Connected Renewable Electricity Generation
 Version No. : Version 18.0; EB 81
 Reference : CDM Methodologies¹²

Tools referred with above methodology and applicable for project activity are:

- Guidelines on the Methodological tool, TOOL21- Demonstration of additionality of small-scale project activities, Version 13.1¹³.
- Tool to calculate the emission factor for an electricity system, TOOL07, Version 07.0 EB 100, Annex 4¹⁴.
- Investment analysis, TOOL27, Version 12.0 EB 116, Annex 2¹⁵.

3.2 Applicability of Methodology

The small-scale grid connected project activity instances under grouped project activity will meet the applicability conditions of the approved methodology AMS I.D, Version 18.0, Sectoral Scope 1, as described below-

Applicability Conditions	Project Applicability
1. This methodology is applicable to project activities that (a) install a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project activity (Greenfield plant); (b) involve a capacity addition in (an) existing plant(s); (c) involve a retrofit of (an) existing plant(s); or (d) involve a replacement of (an) existing plant(s).	The project activity is installation of new power plant at site where there was no renewable energy power plant operating prior to the implementation of the project activity (Greenfield plant). Hence this condition is applicable.
2. Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology:	The project activity is a run of river project activity. Hence this condition is not applicable.

¹² <https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQOH4SBK>

¹³ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-21-v13.1.pdf>

¹⁴ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

¹⁵ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf>

• The project activity is implemented in an existing reservoir with no change in the volume of reservoir; • The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the Project Emissions section, is greater than 4 W/m²; • The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the Project Emissions section, is greater than 4 W/m².	
3. If the new unit has both renewable and non-renewable components (e.g., a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	Not applicable, the project activity is the Renewable Energy Project i.e., Hydro Power Project.
4. Combined heat and power (co-generation) systems are not eligible under this category.	Not applicable, the project activity is the Renewable Energy Project i.e., Hydro Power Project
5. In the case of project activities that involve the addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	The project activity doesn't involve the addition of renewable energy generation units at an existing renewable power generation facility; hence this condition is not applicable.
6. In the case of retrofit or replacement, to qualify as a small-scale project, the total output of the retrofitted or replacement unit shall not exceed the limit of 15 MW.	The project activity doesn't retrofit or replace an existing facility; hence this condition is not applicable.
7. In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid, then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as "AMS-I.C.: Thermal energy production with or without electricity" shall be explored.	The Project activity is a hydro power project and is not a landfill gas, waste gas, waste water treatment and agro-industries projects or recovered methane emissions project. Hence the criteria is not applicable.
8. In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply.	The Project activity is a hydro power project and is not a biomass project. Hence the criteria is not applicable.

Applicability conditions of “Tool to calculate the emission factor for an electricity system”, TOOL07, version 7.0.	
1. This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).	This condition is applicable. OM, BM and CM are estimated using the tool under section 5.1 for calculating baseline emissions.
2. Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off grid power plants. In the latter case, the conditions specified in “Appendix 2: Procedures related to off grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Since the project activity is grid connected, this condition is applicable and the emission factor has been calculated accordingly.
3. In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in Nepal, a non-Annex I country ¹⁶ . Therefore, this criterion is not applicable.
4. Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	The project activity is a grid connected hydro power project. Therefore, this criterion is not applicable.

Applicability conditions of Demonstration of additionality of small-scale project activities, TOOL21, version 13.1.	
1. Para 4- The use of the methodological tool “Demonstration of additionality of small-scale project activities” is not mandatory for project participants when proposing new methodologies. Project participants and coordinating/managing entities may	This condition is applicable. The project participants not proposing new methodology.

¹⁶
https://unfccc.int/process/parties-non-party-stakeholders/parties-convention-and-observer-states?field_national_communications_target_id%5B514%5D=514&field_parties_date_of_ratifi_value=All&field_parties_date_of_signature_value=All&field_parties_date_of_ratifi_value_1=All&field_parties_date_of_signature_value_1=All&combine=

propose alternative methods to demonstrate additionality for consideration by the Executive Board.	
2. Para 5- Project participants and coordinating/managing entities may also apply “TOOL19: Demonstration of additionality of microscale project activities” as applicable.	This condition is not applicable as project is not a microscale project activity.

Applicability conditions of Tool for Investment analysis, TOOL27, version 12.0.

1. Para 2- This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	This condition is applicable. The tool in accordance with the guidance under “Tool for the demonstration and assessment of additionality”.
2. Para 3- In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	This condition is applicable as the tool in accordance with the applied methodology.

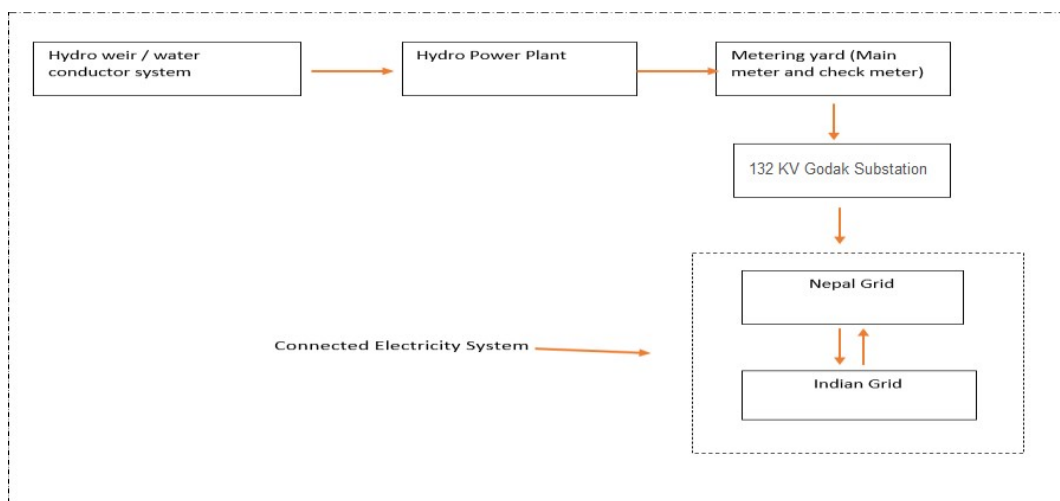
The project activity qualifies as Type I during every year of the crediting period in accordance with applicable provisions for project activity eligibility as discussed above. Also, the total installed capacity of each project activity instance less than 15 MW threshold limit of small-scale project activity instances and respective methodology will be applicable for that project activity instances. The capacity limit is 15 MW for small scale project activity. This project activity is group of number of projects where capacities are varying from each other. However, to meeting applicability criteria of small scale each project activity should not have capacity more than 15 MW.

3.3 Project Boundary

As per AMS-I.D Version 18, EB 81 – “The spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the hydro project power plant is connected to”.

The project boundary is therefore the physical boundary, which includes the project site and all power plants physically connected through transmission and distribution lines to the project activity and that can be dispatched without significant transmission

constraints. The below figure illustrates the boundary of project activity. Therefore, the entire national grid and all connected power plants have been considered in the project boundary for the proposed project activity.



Source		Gas	Included?	Justification/Explanation
Baseline	Grid connected electricity generation	CO ₂	Yes	Main emission source
		CH ₄	NO	Minor emission source
		N ₂ O	NO	Minor emission source
		Other		
Project	Greenfield Hydro Power Project activity	CO ₂	NO	No CO ₂ emissions are emitted from the project
		CH ₄	NO	Project activity does not emit CH ₄
		N ₂ O	NO	Project activity does not emit N ₂ O
		Other		

3.4 Baseline Scenario

As per the applicable methodology, a Greenfield power plant is defined as “a new renewable energy power plant that is constructed and operated at a site where no renewable energy power plant was operated prior to the implementation of the project activity”.

Nepal is interconnected with India at various places through 11kV, 33kV, 132kV and 220kV lines. For transfer of bulk power, interconnection between India and Nepal through Dhalkebar (Nepal) - Muzaffarpur (India) 400kV D/C transmission line has been constructed. A total of about 700 MW of power is being supplied to Nepal through these interconnections. In the regional grid consisting of Nepal and the Indian grid, electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants operational by the addition of new generation sources into regional grid.

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as described in section 5.1.

3.5 Additionality

The additionality of the Project shall be demonstrated by applying the following approach, consisting of two components:

- (i) A Legal Requirement Test; and
 - (ii) An Additionality Test either based on a Positive List test or a projects-specific additionality test.
- (a) Legal Requirement Test

The project is not enforced by law. The project passes the legal requirement test since there are no enforced laws, statutes, regulations, court orders, environmental-mitigation agreements, permitting conditions of other legally-binding mandates requiring its implementation. Since voluntary commitments/agreements within a sector or by an entity do not constitute the legal requirement, the outcome of the legal requirement test is concluded as positive.

(b) Additionality Test

Additionality Assessment

As per the 'TOOL21- Methodological tool: Demonstration of additionality of small-scale project activities Version 13.1 section 5 para 11(a) and Tool 32- Methodological tool: positive list of technologies- Version 04.0., a positive list of grid-connected renewable electricity generation technologies is listed that are automatically defined as additional, without further documentation of barriers. The positive list comprises of the following grid-connected renewable electricity generation technologies of installed capacity up to 15 MW.

a) Tool 21

- (i) Solar technologies (photovoltaic and solar thermal electricity generation); (ii) Off-shore wind technologies; (iii) Marine technologies (wave, tidal); (iv) Building-integrated wind turbines or household rooftop wind turbines of a size up to 100 kW;

b) Tool 32

a) Solar thermal electricity generation including concentrating solar power; (b) Off-shore wind technologies; (c) Marine wave technologies; (d) Marine tidal technologies; (e) Building-integrated wind turbines or household rooftop wind turbines of a size up to 100 kW; (f) Biomass internal gasification combined cycle.

As per the above criteria the grouped project activity is not identified as automatically additional.

As per Guidelines on the Demonstration of Additionality of Small-scale Project Activities (Ver. 13.1 EB 105 Annex 4)¹⁷, to establish the project additionality, it has to be shown that the project activity would not have occurred anyway due to at least one of the following barriers:

- **Investment barrier:** a financially more viable alternative to the project activity would have led to higher emissions;
- **Technological barrier:** a less technologically advanced alternative to the project activity involves lower risks due to the performance uncertainty or low market share of the new technology adopted for the project activity and so would have led to higher emissions;
- **Barrier analysis:** prevailing practice or existing regulatory or policy requirements would have led to implementation of a technology with higher emissions;
- **Other barriers:** without the project activity, for another specific reason identified by the project Owner, such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies, emissions would have been higher.

The project investor has selected Investment barrier to demonstrate in a conservative and transparent manner that the proposed project activity is financially unattractive.

Appropriateness of using benchmark analysis for additionality demonstration and its conformity to Investment Analysis Annex 2, EB 116¹⁸ -

Considering the fact that the alternative to the project is the supply of electricity from the grid & the choice of the developer is to invest or not to invest, benchmark analysis has been considered appropriate for demonstration of additionality, which is in conformity with “Investment Analysis” Annex 2 EB 116.

Selection of Benchmark & Financial Indicator:

According to the “Demonstration of additionality of small-scale project activities¹⁹”, the financial indicator can be based either on (1) project IRR or (2) equity IRR. There is no general preference

¹⁷ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-21-v13.1.pdf>

¹⁸ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf>

¹⁹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-21-v13.1.pdf>

between the approaches (1) or (2). The benchmark chosen for analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for project activity has been chosen as post tax equity IRR.

Determine appropriate analysis method

As per Sub-step 2a, Paragraph (1), as the project activity is selling the generated electricity to grid & getting financial benefits other than VCS benefits, hence, Option- I (Apply simple cost analysis) is not applicable under this situation. Also, as per Investment analysis EB-116²⁰, Annex 2, “If the alternative to the project activity is the supply of electricity from a grid this is not to be considered an investment and a benchmark approach is considered appropriate”. Hence, Option-II (Apply investment comparison analysis) is also not applicable under this situation. So, the project promoter has chosen Option- III or benchmark analysis as an appropriate analysis method to demonstrate the investment barrier.

Benchmark Calculation

At the time of decision made of project activity, Version 12 of methodological tool "Investment Analysis" (EB 116, Annex 2) was the latest available tool to Project owner. Hence, Project owner has used Methodological Tool for Investment Analysis version 12 (EB 116²¹, Annex 02) which is appropriate and more conservative for benchmark calculation and Project owner has considered the same tool for default value of return on equity for the project. The default value of Return on Equity for Group-1 projects in Nepal is 12.62% as per EB 116, Annex 02.

Chronology Table:

Sr. No.	Investor Name	Investment Decision Date	Commissioning Date
1.	Samling Power Company Ltd.	18-March-2019	17-September-2021
2.	Asian Hydropower Ltd.	18-March-2019	01-November-2021

The benchmark has been considered in accordance with EB 116 Annex 2, “The values in the table in Appendix A may also be used, as a simple default option”.

²⁰ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf>

²¹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf>

As suggested in Appendix A, in EB 116 Annex 2 of the “investment analysis” version 12.0, the default value benchmark of expected return on equity, in real terms for Energy Industries (Group 1) in Nepal = 12.62%.

Methodology deployed for arriving at a suitable value of Benchmark using Default Value has been described below:

- As the proposed project activity generates power utilizing hydro energy, Group 1 as per para 5 of Appendix of EB 116, Annex 2 has been identified as a suitable category.
- The investment analysis has been carried out in Nominal terms. Accordingly, default value as given in Para 16, Annex 2, EB 116 has been adjusted by adding suitable forecasted inflation rate taken from IMF.
- Project investor has calculated Benchmark based on WPI median inflation rate. As per Para 16 of Appendix of EB 116, Annex 2, the inflation forecast should be for the duration of the crediting period. However, since RBI provides forecast inflation only for 5 & 10 years, the project investor has calculated benchmark using inflation forecast (WPI Mean) for 10 years as the most conservative value for the project activity.

The benchmark has been computed in the following manner:

$$\text{Nominal Benchmark}^{22} = \{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$$

Where,

Real Benchmark = Default Value, i.e., 12.62% (as per Appendix of Annex 2, EB 116)

Inflation rate = Projected Inflation Rate for Nepal Default Value Benchmark:

Default Value Benchmark:

For Samling Power Company Ltd.

Benchmark Calculations	Value	Sources Link	Document Date
Default Value for Nepal as per UNFCCC guidelines	12.62%	https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf	02-November-2022
Inflation forecast (WPI Mean) as per IMF for 10yrs	4.60%	https://www.imf.org/en/Countries/NPL	01-March-2019
Benchmark (with 10yrs Forecast)	17.80%	Calculated	
Benchmark	17.80%	Calculated	

²² As per Pg. 320 of Corporate Finance, Second Edition of Aswath Damodaran

For Asian Hydropower Ltd.

Benchmark Calculations	Value	Sources Link	Document Date
Default Value for Nepal as per UNFCCC guidelines	12.62%	https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-27-v12.pdf	02-November-2022
Inflation forecast (WPI Mean) as per IMF for 10yrs	4.60%	https://www.imf.org/en/Countries/NPL	01-March-2019
Benchmark (with 10yrs Forecast)	17.80%	Calculated	
Benchmark	17.80%	Calculated	

Calculation and comparison of financial indicators

Key Assumptions supporting financial projections are provided in excel spreadsheet to the VVB. Based on result of IRR excel spreadsheets, equity IRR is less than Benchmark. The input assumption and the IRR outcome can be referred in below:

For Samling Power Company Ltd.

Details of the project		Source	Link
State where the project is situated	Ilam District Province 1	As per DPR, page no. 15	
Total Capacity (MW)	9.51	As per DPR, page no. 15	
Expected Date of Commissioning	17-Sep-21		
Life of the plant (Yrs.)	30	As per DPR, page no. 105	-
Generation of electricity			
PLF (%)	54.51%	As per DPR, page no. 96	-
Annual generation (kWh)	4,54,10,973	Calculated Value	

Auxiliary consumption	0.00%		-
Tariff rate at the decision making (INR/kWh)	2.45	As per DPR, page no. 99	-
Escalation in tariff rate	0.0%		-
Transmission & Wheeling Losses (%)	0.00%		-
Operation and maintenance cost and Insurance			
O & M Expenses (INR Mn.)	11.87	As per DPR, page no. 106	-
O & M free for (Yr.)	-		
Escalation in the operational expenses (%)	3.00%	As per DPR, page no. 106	-
Insurance (INR Mn.)	3.56	Calculated Value	-
Financial parameters			
TOTAL COST (INR Mn.)	1,187.90	As Per DPR, page no. 104	
Loan Amount (INR Mn.)	890.92	As Per DPR, page no. 105	
Equity Investment (INR Mn.)	296.97	Calculated Value	
Term loan			
Loan Amount (INR Mn.)	890.92	As Per DPR, page no. 105	-
Interest rate (%)	11.50%	As Per DPR, page no. 105	-

Loan Tenure (Qtr.)	40	As Per DPR, page no. 107	-
Moratorium Period (Qtr.)	-	Not Applicable	
Repayment Period (Qtr.)	40	Calculated Value	-
Repayment instalments value (INR Mn.)	22.273	Calculated Value	-
1st instalment from (Qtr. end)	31-Dec-21	Considered from the next Quarter End	-
Book Depreciation (SLM Method)			
Land	-		
Gross Depreciable Value (INR Mn.)	1,187.90	Calculated Value	
Salvage Value (%)	10.00%	As per RERC order dt. 23.01.2009, Pg 20	
Salvage value (INR Mn.)	118.79	Calculated Value	-
Net Depreciable Value (INR Mn.)	1,069.11	Calculated Value	
Residual Value (INR Mn.)	118.79	Calculated Value	
IT Depreciation			-
IT Depreciation (%)	30.00%	IT act	
Income Tax			
Financial Year	FY 2019-20		

Income tax rate (%)	20.00%	As Per Income Tax Rule	
Corporate Tax / MAT (%)	25.00%	As Per IT rule	https://www.orbitax.com/taxhub/countrychapters/NP/Nepal/7890123caa2f4bbc950c93677678bece/Corporate-Income-Tax-588#:~:text=The%20standard%20corporate%20income%20tax,income%20tax%20i.e.%20at%2025%25.
GST (%)	13.00%	As Per Income Tax Rule	https://tradingeconomics.com/nepal/sales-tax-rate
Surcharge (%)	20.00%	As Per Income Tax Rule	https://www.nrb.org.np/contents/uploads/2020/09/Annual-Report-2018.19.pdf
Education cess (%)	4.00%	As Per Income Tax Rule	
Final Tax rates			
Income tax rate (%)	24.96%	Calculated Value	
MAT (%)	31.20%	Calculated Value	
GST (%)	13.52%	Calculated Value	

For Asian Hydropower Ltd.

Details of the project		Source	Link
State where the project is situated	Ilam District Province 1	As per DPR, page no. 5	
Total Capacity (MW)	6.20	As per DPR, page no. 6	

Expected Date of Commissioning	1-Nov-21		
Life of the plant (Yrs.)	30	As per DPR, page no. 128	-
Generation of electricity			
PLF (%)	54.51 %	As per DPR, page no. 110	-
Annual generation (kWh)	2,96,05,471	Calculated Value	
Auxiliary consumption	0.00%		-
Tariff rate at the decision making (INR/kWh)	2.45	As per DPR, page no. 123	-
Escalation in tariff rate	0.0%		-
Transmission & Wheeling Losses (%)	0.00%		-
Operation and maintenance cost and Insurance			
O & M Expenses (INR Mn.)	7.39	As per DPR, page no. 128	-
O & M free for (Yr.)	-		
Escalation in the operational expenses (%)	3.00%	As per DPR, page no. 65	-
Insurance (INR Mn.)	2.22	Calculated Value	-
Financial parameters			
TOTAL COST (INR Mn.)	739.14	As Per DPR, page no. 125	
Loan Amount (INR Mn.)	554.36	As Per DPR, page no. 129	

Equity Investment (INR Mn.)	184.79	Calculated Value	
Term loan			
Loan Amount (INR Mn.)	554.36	As Per DPR, page no. 129	-
Interest rate (%)	11.50 %	As Per DPR, page no. 129	-
Loan Tenure (Qtr.)	40	As Per DPR, page no. 129	-
Moratorium Period (Qtr.)	-	Not Applicable	
Repayment Period (Qtr.)	40	Calculated Value	-
Repayment instalments (INR Mn.) value	13.859	Calculated Value	-
1st instalment from (Qtr. end)	31-Mar-22	Considered from the next Quarter End	-
Book Depreciation (SLM Method)			
Land	-		
Gross Depreciable Value (INR Mn.)	739.14	Calculated Value	
Salvage Value (%)	10.00 %	As per RERC order dt. 23.01.2009, Pg 20	
Salvage value (INR Mn.)	73.91	Calculated Value	-

Net Depreciable Value (INR Mn.)	665.23	Calculated Value	
Residual Value (INR Mn.)	73.91	Calculated Value	
IT Depreciation			-
IT Depreciation (%)	30.00 %	IT act	
Income Tax			
Financial Year	FY 2019-20		
Income tax rate (%)	20.00 %	As Per Income Tax Rule	
Corporate Tax / MAT (%)	25.00 %	As Per IT rule	https://www.orbitax.com/taxhub/countrychapters/NP/Nepal/7890123caa2f4bbc950c93677678bece/Corporate-Income-Tax-588#:~:text=The%20standard%20corporate%20income%20tax,income%20tax%20i.e.%20at%2025%25.
GST (%)	13.00 %	As Per Income Tax Rule	https://tradingeconomics.com/nepal/sales-tax-rate
Surcharge (%)	20.00 %	As Per Income Tax Rule	https://www.nrb.org.np/contents/uploads/2020/09/Annual-Report-2018.19.pdf
Education cess (%)	4.00%	As Per Income Tax Rule	
Final Tax rates			
Income tax rate (%)	24.96 %	Calculated Value	
MAT (%)	31.20 %	Calculated Value	
GST (%)	13.52 %	Calculated Value	

Sensitivity Analysis

The results of sensitivity analysis are as follows:

For Samling Power Company Ltd.

Sensitivity Analysis	Equity IRR			
Variation %	-10%	Normal	10%	Breaching Value
PLF	1.63%	3.21%	4.77%	100.20%
O&M	3.47%	3.21%	2.96%	-728.50%
Project Cost	4.67%	3.21%	2.01%	-53.75%
Tariff Rate	1.63%	3.21%	4.77%	100.20%

For Asain Hydropower Ltd.

Sensitivity Analysis	Equity IRR			
Variation %	-10%	Normal	10%	Breaching Value
PLF	2.14%	3.74%	5.31%	95.38%
O&M	3.97%	3.74%	3.50%	-727.00%
Project Cost	5.22%	3.74%	2.51%	-52.33%
Tariff Rate	2.14%	3.74%	5.31%	95.38%

As per Guidance 27 and 28 of Investment Analysis Annex 2 of EB 116, only variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation and the results of this variation should be presented in the PDMR and be reproducible in the associated spreadsheets. Guidance also states, “All parameters varied need not necessarily be subjected to both negative and positive variations of the same magnitude”. The Annex also states, as a general point of departure, variations in the sensitivity analysis should at least cover a range of +10% and –10%, unless this is not deemed appropriate in the context of the specific project circumstances.

Since the project cost is already firmed up, the cost is not variable. The tariff is determined by PPA which is fixed for years mentioned as per the respective State Electricity Board’s tariff order and hence it need not be subjected to variation. All other expenses are much less than 20% of the total cost. Hence, only PLF needs to be subjected to reasonable variation. Nevertheless, following factors have been subjected to sensitivity analysis:

1. PLF
2. O&M Cost
3. Project Cost
4. Tariff

The results of sensitivity analysis show that the even with a variation of +10% & -10% in the PLF, O&M cost, Project cost and Tariff Rate, Equity IRR is significantly lower than the benchmark. And

it is evident from the results given above; the project remains additional even under the most favorable conditions.

1. Samling Power Company Ltd.

Probability to breach the benchmark:
Sensitivity Parameter 1: PLF or generation
PLF considered in financials is as per Third Party report in line with “Guidelines for the reporting and validation of Plant load factors” stated in EB 48 Annex11 option 3(b). Hence, variation in PLF of more than 10% is unlikely to happen as the PLF has been reported as per the Third-Party report based on long term data. IRR is not crossing the benchmark if PLF increases more that 10% in any circumstances. IRR is crossing the benchmark if PLF increased be more than 100.20%. Project is already operational and actual PLF is higher and still below the benchmark. Hence project is still additional.
Sensitivity Parameter 2: O&M
The sensitivity analysis reveals that O&M will breach the benchmark at negative values and is hypothetical case. Since the O&M cost is generally subject to escalation and also subject to inflationary pressure, any reduction in the O&M costs is highly unlikely. Hence, the reduction in the O&M cost is highly unlikely. IRR is crossing the benchmark if O&M cost is reduced by -728.50%. O&M agreement is already in place by the PP and O&M is actual less than what was estimated while calculating IRR. But still the IRR is 4.19% which is well below the benchmark. Hence project is still additional.
Sensitivity Parameter 3: Project Cost
Project Cost for financial analysis is considered based on the information of DPR, being available at the time of investment decision making to go ahead with the project activity. However, the Sensitivity is carried out for 10% variation and for threshold level below which benchmark is not breached. Thus, it is unlikely that project cost will change beyond sensitivity range. IRR will cross the benchmark if project cost is reduced by -53.75%. Project is already operational and actual project cost is higher. Hence project is still additional.
Sensitivity Parameter 4: Tariff Rate
The tariff is fixed for entire lifetime of the project activity. Hence, there is no probability to get variation for the same. However, sensitivity is carried out for +/-10% even then the benchmark is not breached. Hence variation is not a possible scenario.

2. Asain Hydropower Ltd.

Probability to breach the benchmark:
Sensitivity Parameter 1: PLF or generation
PLF considered in financials is as per Third Party report in line with “Guidelines for the reporting and validation of Plant load factors” stated in EB 48 Annex11 option 3(b). Hence, variation in PLF of more than 10% is unlikely to happen as the PLF has been reported as per the Third-Party report based on long term data. IRR is not crossing the benchmark if PLF increases more that 10% in any circumstances. IRR is crossing the benchmark if PLF increased be more than 95.38%. Project is already operational and actual PLF is higher and still below the benchmark. Hence project is still additional.
Sensitivity Parameter 2: O&M
The sensitivity analysis reveals that O&M will breach the benchmark at negative values and is hypothetical case. Since the O&M cost is generally subject to escalation and also subject to inflationary

pressure, any reduction in the O&M costs is highly unlikely. Hence, the reduction in the O&M cost is highly unlikely. IRR is crossing the benchmark if O&M cost is reduced by -727.00%. O&M agreement is already in place by the PP and O&M is actual less than what was estimated while calculating IRR. But still the IRR is 3.94% which is well below the benchmark. Hence project is still additional.

Sensitivity Parameter 3: Project Cost

Project Cost for financial analysis is considered based on the information of DPR, being available at the time of investment decision making to go ahead with the project activity. However, the Sensitivity is carried out for 10% variation and for threshold level below which benchmark is not breached. Thus, it is unlikely that project cost will change beyond sensitivity range. IRR will cross the benchmark if project cost is reduced by -52.33%. Project is already operational and actual project cost is less. But still the IRR is 5.4% which is well below the benchmark. Hence project is still additional.

Sensitivity Parameter 4: Tariff Rate

The tariff is fixed for entire lifetime of the project activity. Hence, there is no probability to get variation for the same. However, sensitivity is carried out for +/-10% even then the benchmark is not breached. Hence variation is not a possible scenario.

The result of the analysis is as follows:

This substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark Equity IRR) for any of the investor. Thus, it can be easily concluded that project activity is additional & is not business as usual scenario. As per Tool 27 para 16²³ inflation data is not published by Central bank of country. So, PP has considered the IMF (International Monetary Fund World Economic Outlook) data for benchmark.

Sr. No.		Equity IRR	Benchmark (Equity IRR)
1	Samling Power Company Ltd.	3.21%	17.80%
2	Asain Hydropower Ltd.	3.74%	17.80%

Step 3: Barrier analysis

Barrier analysis has not been used.

Step 4: Common practice analysis

Not applicable as it is a small-scale project.

²³ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf#page=6>

Conclusion: As described above, the project fulfils all necessary requirements of additionality specified in the 'Demonstration of additionality of small-scale project activities' v13.1. Hence, the project is additional.

3.6 Methodology Deviations

Not Applicable

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

The project activity involves installation and generation of electricity using renewable energy resources i.e., by using the hydro power (renewable sources) replacing electricity supply from a fossil-fuel dominated electricity, thus leads to reductions of anthropogenic GHG emissions from atmosphere.

The electricity generated is supplied to Integrated Nepal Power System (INPS) through 4 km long, 33 kV Single Circuit Transmission line from the project to the 132/33kV Godak sub-station of NEA. Emission factor of Nepal electricity system is not available as it is not published by Nepal Electricity Authority (NEA). Hence PP has taken the emission factor of Nepal is considered as zero from IFI Default Grid Factors 2021 v3.2 published by UNFCCC²⁴. However, Nepal is importing electricity from India. Approximately 32% electricity is imported from Indian grid and emission factor of Indian grid is calculated as 0.9105 by using CEA database²⁵ as per TOOL07. Hence the emission factor of Nepal is calculated as weighted average emission factor of Nepal and India. Asian Development Bank has published the emission factor of Nepal is 348 tCO₂/ GWh²⁶. Weighted average of Nepal and India grid is taken as 0.2898 tCO₂/ MWh (i.e. 289 tCO₂/ GWh) which is lower than Asian Development Bank and should be considered as a conservative. The operational life time of project activity is 30 years.

Moreover, the operation of the project activity(s) during this monitoring period, including any information events that may impact GHG emission reduction or removals and monitoring is not applicable. And no breakdown occurred in this plant during the verification period of the project activity.

There are no such incidents in the current monitoring period that may impact the GHG emission reductions or removals and monitoring.

Organizational Structure for Monitoring

²⁴ <https://unfccc.int/documents/461676>

²⁵ https://cea.nic.in/wp-content/uploads/baseline/2023/01/Approved_report_emission_2021_22.pdf

²⁶ <https://www.adb.org/sites/default/files/linked-documents/49086-001-crf.pdf>

Responsibilities of project Head: Tracking and reviewing the overall functioning and maintenance of the project activity throughout the areas.

Responsibilities of O & M Head: O & M Manager and team is responsible for Operations and Maintenance related issues; they are also responsible for day-to-day data collection and monitoring.

Technical Specification for Sampling Power Company Ltd.

1. Turbine
 - Type: Horizontal shaft Francis turbine
 - Make: TPSC-I
 - Number: 2
 - Rated output capacity per unit: 4930 KW
 - Net Head: 136.64 m
 - Efficiency (max): 92.8% at minimum head
 - Turbine axis level: 495.58 masl
 - Rated Speed: 750 RPM
 - Runaway speed: 1380 RPM
2. Main Inlet Valve:
 - Make: Fouress Engineering (India) Ltd.
 - Type: Butterfly valve
 - Size: 900 mm
 - Design Pressure: 19.83 Kg/cm²
 - Maximum flow: 4.471 m³/sec
 - Method of Closing: Counter weight
 - Method of opening: Oil pressure
 - Valve Operating time:
 - 1) Normal closing time: 45-60 (adjustable)
 - 2) Normal opening time: 45-60 (adjustable)
3. Generator:
 - Type: Cylindrical Pole, Synchronous Generator
 - Make: TDPS
 - Rated Output capacity per unit: 5597 KVA + 10% COL
 - Power factor: 0.85
 - Voltage: 6.6 KV
 - Frequency: 50 Hz
 - No of units: 2
 - Excitation system: Brushless excitation
 - Efficiency: 96.8 % at 100% of full load 0.85 PF
 - Speed: 750 RPM
4. Oil Pressure Unit (OPU):
 - Make: Hydac (India) Pvt. Ltd.
 - Working Pressure: 120 Bar.
 - Test Pressure: 180 Bar
 - No of Pump: 2 (One working and another in Standby)
 - Rating of Pump: 2.2 KW, 400 V, 50 Hz, 1500 RPM
 - Main Pump Flow: 5 lpm
 - Piping material: Carbon Steel

- Oil reservoir Capacity: 250 Liters
- Working Fluid: Mineral Oil ISO VG 46
- Viscosity of Oil @ 40 Deg. C: 46 cst
- Operating maximum temperature: 80 deg
- 5. Lube oil system (LOS):
 - Make: Pascal System technik Pvt. Ltd.
 - No of pump: 2 AC & 1 DC
 - Rating of AC Pump: 3 HP, 400 V AC, 1445 RPM
 - Rating of DC Pump: 3 HP, 110 V DC, 1500 RPM
 - NDE flow rate: 7 LPM
 - DE Flow rate: 57 LPM
 - Oil Tank Capacity: 630 Liters
- 6. DC System:

Battery Charger

 - Make: Universal Instruments Mfg. Co. Pvt. Ltd.
 - Capacity: 110 V, 250 Ah
 - No of Charger: 2 (1 main & 1 standby) FCBC
 - Float Voltage per cell: 2.23 +/- 0.01 Volt at 25-34 Deg. C temperature
 - Battery Type: VRLA Battery
 - Battery make: EXIDE Industries Ltd.
 - Rating of battery: 54 * 2 V / 250Ah

Technical Specification for Asian Hydropower Ltd.

1. Turbine
 - Type: Horizontal shaft Francis turbine
 - Make: Flovel Energy Private Limited, India
 - Number: 2
 - Rated output capacity per unit: 3213 KW
 - Net Head: 125.49 m
 - Rated Discharge: 2.86 m³/s per unit
 - Efficiency (max): 93.2%
 - Turbine axis level: 514.24 masl
 - Rated Speed: 750 RPM
 - Runaway speed: 1215 RPM
2. Main Inlet Valve:
 - Make: TB Hydro Flovel Valves Private Limited, India.
 - Type: Butterfly valve
 - Size: 900 mm
 - Design Pressure: 18 bar
 - Method of Closing: Counter weight
 - Method of opening: Oil pressure
 - Number of seals: 1 No.
 - Valve Operating time:

Normal closing time: 45-60 (adjustable)

Normal opening time: 45-60 (adjustable)

3. Generator:

- Type: Cylindrical Pole, Synchronous Generator
- Make: WEG Industries Pvt. Ltd., India
- Rated Output capacity per unit: 3647 kVA + 10% COL
- Power factor: 0.85
- Voltage: 6.6 KV
- Frequency: 50 Hz
- No of units: 2
- Excitation system: Brushless excitation
- Efficiency: 96.6 % at 100% of full load 0.85 PF
- Speed: 750 RPM

4. Oil Pressure Unit (OPU):

- Make: Hydac (India) Pvt. Ltd.
- Working Pressure: 120 Bar.
- Test Pressure: 180 Bar
- No of Pump: 2 (One working and another in Standby)
- Rating of Pump: 3.7 kW, 400 V, 50 Hz, 1440 RPM
- Main Pump Flow: 8 lpm
- Piping material: Carbon Steel
- Oil reservoir Capacity: 200 Liters
- Working Fluid: Mineral Oil ISO VG 46/68
- Operating maximum temperature: 80 deg

5. Lube oil system (LOS):

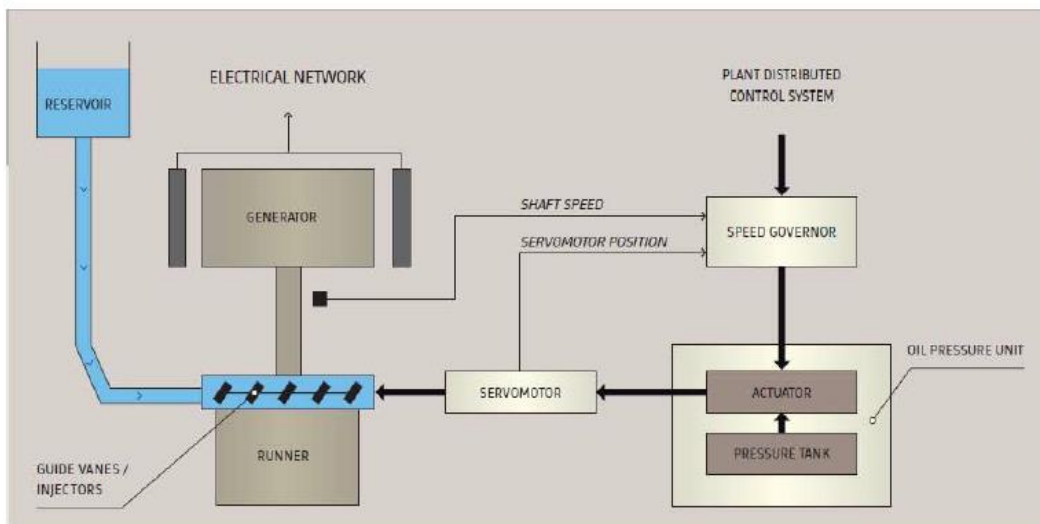
- Make: Dropco Multilub Systems Pvt. Ltd.
- No of pump: 2 AC & 1 DC
- Rating of AC Pump: 1.1 kW, 400 V AC, 1450 RPM
- Rating of DC Pump: 1.1 kW, 110 V DC, 1450 RPM
- NDE flow rate: 4 LPM
- DE Flow rate: 28 LPM
- Oil Tank Capacity: 400 Liters

6. DC System:

Battery Charger

- Make: Green Secure Energy Systems
- Capacity: 110 V, 250 Ah
- No of Charger: 2 (1 main & 1 standby) FCBC, SMPS based technology
- Float Voltage per cell: 2.25 Volt/cell
- Boost Voltage per cell: 2.3 Volt/cell
- Battery Type: VRLA Battery
- Battery make: HBL Power Systems Ltd.
- Rating of battery: 55 * 2 V / 250Ah

The main equipment diagram of the hydro power plant:



5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The project activity mainly reduces carbon dioxide through substitution of grid electricity generation with fossil fuel fired power plants by renewable electricity. As per para 43 of the methodology AMS I D Version 18,²⁷ the emission reduction ER_y by the project activity during a given year y is the difference between baseline emissions (BE_y), project emissions (PE_y) and emissions due to leakage (LE_y), as follows:

$$ER_y = BE_y - PE_y - LE_y$$

where:

ER_y = Emission reductions in year y (t CO₂/y)

BE_y = Baseline Emissions in year y (t CO₂/y)

PE_y = Project emissions in year y (t CO₂/y)

LE_y = Leakage emissions in year y (t CO₂/y)

Baseline Emissions:

Baseline emissions in the project activity is calculated as per para-22 of AMS I.D. version-18:

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

where:

²⁷ <https://cdm.unfccc.int/UserManagement/FileStorage/2P7FS6ZQAR84LG3NMKYUH50WI90DBC>

BE_y	= Baseline emissions in year y (t CO ₂)
$EG_{PJ,y}$	= Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in project year (MWh)
$EF_{grid,y}$	= Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO ₂ /MWh)

• Calculation of the CO₂ emission factor of the grid

As per “Tool to calculate the Emission Factor for an electricity system version-07.0”, following steps has to be applied:

- (a) Step 1: Identify the relevant electricity systems;
- (b) Step 2: Choose whether to include off-grid power plants in the project electricity system (optional);
- (c) Step 3: Select a method to determine the operating margin (OM);
- (d) Step 4: Calculate the operating margin emission factor according to the selected method;
- (e) Step 5: Calculate the build margin (BM) emission factor;
- (f) Step 6: Calculate the combined margin (CM) emission factor

Step 1: Identify the relevant electricity systems

Nepal is interconnected with India at various places through 11kV, 33kV, 132kV and 220kV lines²⁸. For transfer of bulk power, interconnection between India and Nepal through Dhalkebar (Nepal) - Muzaffarpur (India) 400kV D/C transmission line has been constructed. A total of about 7008 MW of power is being supplied to Nepal through these interconnections²⁹.

Further, 400kV D/C Gorakhpur (India) – Butwal (Nepal) line, 400kV D/C Dhalkebar (Nepal) – Sitamarhi (India) line, 132kV D/C Nanpara, Bihar (India) – Kohalpur (Nepal), stringing of second circuit of 132kV line Kataiya (India) – Kushaha (Nepal) and 132 kV Raxaul (India) – Parwanipur (Nepal) lines have been agreed. Information has been taken from Annual report 2021/22 database of Nepal Electricity Authority.

As per Para 15 and 16 of ‘Tool to calculate the Emission Factor for an electricity system version-07.0’ (Tool 07), Project owner has identified Project electricity system for determining the electricity emission factors and connected electricity systems as Nepal and Indian+Nepal Grid respectively.

As per para 17 of ‘Tool to calculate the Emission Factor for an electricity system version-07.0’ (Tool 7),

Project Owner has chosen the ‘Option 3’: A delineation of the project electricity system defined by more than one independent dispatch areas, e.g., multi-national power pools.

²⁸ <https://powermin.gov.in/en/content/interconnection-neighbouring-countries>

²⁹ https://www.nea.org.np/annual_report

As per para 18 of 'Tool to calculate the Emission Factor for an electricity system version-07.0' (Tool 7): In case of option 3, transmission lines between dispatch areas included in the proposed delineation shall be checked for the existence or non-existence of transmission constraints.

As per para 19 of the 'Tool to calculate the Emission Factor for an electricity system version-07.0' (Tool 7):

There are no transmission constraints if any one of the following criteria is met:

(b) In case of electricity systems with spot markets for electricity: there are differences in electricity prices (without transmission and distribution costs) of less than five per cent between the two electricity systems at least during 90 per cent of the hours of the most recent year for which information is available (at least one year data is required); or

(c) The transmission line(s) is operated at 75 per cent or less of its rated capacity during 90 per cent or more of the hours of the most recent year for which information is available (at least one year data is required) using the algorithm below:

- (i) For every hour of the year check whether the transmission line is operated at 75 per cent or less of its rated capacity;
- (ii) Each hour of the year when the transmission line was operated at 75 per cent or less of its rated capacity should be counted as zero;
- (iii) Each hour of the year when the transmission line was operated at 75 per cent or more of its rated capacity should be counted as one;
- (iv) There is no transmission constraint if the total sum is less than ten per cent of the hours of the year (e.g., 876 for even year and 878 for leap year);
- (v) The algorithm can be illustrated by the following equation:

$$\sum_{1}^{8760} \left[\frac{\text{Hourly power transmission (MWh)}}{[\text{Maximum line's load capacity (MW)}]} > 75\text{per cent} \right] < 876$$

- (vi) The maximum line's load capacity should be based on official information (e.g., from the operator of the system);

(d) The transmission capacity of the transmission line(s) that is connecting electricity systems is more than 10 per cent of the installed capacity either of the project electricity system or of the connected electricity system, whichever is smaller.

Project Owner has shown below that there is no transmission constraint according to para 19 (c): Transmission capacity of the transmission line(s) that is connecting electricity systems (Indian grid to Nepal Grid) = 700 MW

Installed Capacity of the project electricity system (Nepal³⁰) = 1451.34 MW Installed Capacity of connected electricity system (India³¹) = 3,95,075 MW

³⁰ https://www.nea.org.np/annual_report

³¹ <https://powermin.gov.in/en/content/power-sector-glance-all-india>

Hence, 10 % of the project electricity system is 145.13 MW. And thus, the transmission capacity of the transmission line(s) that is connecting electricity systems (700 MW) is more than 10 percent of the installed capacity (145.13 MW)³².

As per Para 25 of 'Tool to calculate the Emission Factor for an electricity system version-07.0' (Tool 7): Project owner has chosen option b as mentioned in tool to calculate the simple operating margin emission rate of the exporting grid (in this project activity Indian grid is the exporting grid). And details have mentioned in Step 3.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)

Project participants may choose between the following two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

As the project is exporting electricity to the NEWNE regional grid, thus, the project participants have chosen Option I for the calculation of the operating and build margin emission factor i.e., off grid power plants are not being included in the calculation.

Step 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods:

- (a) Simple OM, or
- (b) Simple adjusted OM, or
- (c) Dispatch data analysis OM, or
- (d) Average OM.

The data required to calculate Simple adjusted OM and Dispatch data analysis OM is not possible due to lack of availability of data to project developers. The choice of other two options for calculating operating margin emission factor depends on generation of electricity from low-cost/ must –run sources. In the context of the methodology low cost/must run resources typically include hydro, geothermal, wind, low-cost biomass, nuclear and solar generation.

Share of Low-cost Must-Run (Hydro/Nuclear) (% of Net generation)

	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22
India	14.6%	14.3%	14.5%	17.0%	16.5%	15.8%

Data Source: Central Electricity Authority (CEA) database Version 18.0, October 2022

The above data clearly shows that the percentage of total grid generation by low-cost/ must-run plants (on the basis of average of five most recent years) for the Indian grid is less than 50 % of the total generation. Thus, the Average OM method cannot be applied, as low cost/must run resources constitute less than 50% of total grid generation.

³² https://www.nea.org.np/annual_report

The Simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (t CO₂/MWh) of all generating power plants serving this system, not including low-cost/must-run power plants/units.

For the simple OM, the emissions factor can be calculated using either of the following data vintages.

a) Ex-ante options: If the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emission factor during the crediting period is required.

Or

b) Ex-post option: If the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring.

PP has chosen ex-ante option for calculation of Simple OM emission factor using a 3 years' generation-weighted average, based on the most recent data available at the time of submission.

OM determined at validation stage will be the same throughout the crediting period. There will be no requirement to monitor & recalculate the emission factor during the crediting period.

Step 4: Calculate the operating margin emission factor ($EF_{grid, OMSimple,y}$) according to the selected method

The operating margin emission factor has been calculated using a 3-year data vintage:

Net Generation in operating Margin (GWh) (Incl. Imports)			
	2019-20	2020-21	2021-22
Indian Grid (Connected System)	965,009	958,218	1,035,672
Simple Operating Margin (t CO ₂ /MWh) (incl. imports)			
	2019-20	2020-21	2021-22
Indian Grid (Connected System)	0.9555	0.9405	0.9605
Weighted Generation Operating Margin Indian Grid (Connected System)		0.9524	

Step 5: Calculate the build margin (BM) emission factor ($EF_{grid,BM,y}$)

As per CDM Methodological tool" Tool to calculate the emission factor for an electricity system" (Version 07.0, EB 100, Annex 4) para 72:

In terms of vintage of data, project owner can choose between one of the following two options:

a) **Option 1-** for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of VCS Joint PDMR submission to the VVB. For the second crediting period, the build margin

emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the VVB. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

- b) **Option 2-** For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex-ante as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

Note: In VCS there is fixed crediting period for 10 years, no renewal (2nd or 3rd) of crediting period will be there. So, BM is fixed for the entire crediting period.

Option 1 as described above is chosen by PP to calculate the build margin emission factor for the project activity. BM is calculated ex-ante based on the most recent information available at the time of submission of VCS Joint PDMR and is fixed for the entire crediting period.

Build Margin (tCO ₂ /MWh) (not adjusted for imports)	
Year	2020-21
Indian Grid	0.8687

Step 6: Calculate the combined margin (CM) emission factor (EF_{grid,CM,y})

As per CDM Methodological tool “Tool to calculate the emission factor for an electricity system” (Version 07.0, EB 100, Annex 4) para 81: -

The calculation of the combined margin (CM) emission factor (EF_{grid,CM,y}) is based on one of the following methods:

- a) Weighted average CM: or
b) Simplified CM

PP has chosen option (a) i.e., weighted average CM to calculate the combined margin emission factor for the project activity.

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} * W_{OM} + EF_{grid,BM,y} * W_{BM}$$

Where;

EF _{grid,CM,y}	Build margin CO ₂ emission factor in year y (t CO ₂ / MWh)
EF _{grid,OM,y}	Operating margin CO ₂ emission factor in year y (t CO ₂ /MWh)
W _{OM}	Weighting of operating margin emissions factor (per cent)
W _{BM}	Weighting of build margin emissions factor (per cent)

The following default values should be used for W_{OM} and W_{BM} as per Para 86 of Tool 7:

a) Wind and solar power generation project activities: $w_{OM} = 0.75$ and $w_{BM} = 0.25$ (owing to their intermittent and non-dispatchable nature) for the first crediting period and for subsequent crediting periods;

b) All other projects: $w_{OM} = 0.5$ and $w_{BM} = 0.5$ for the first crediting period, and $w_{OM} = 0.25$ and $w_{BM} = 0.75$ for the second and third crediting period,⁸ unless otherwise specified in the approved methodology which refers to this tool.

For Hydro project activities $W_{OM} = 0.5$ and $W_{BM} = 0.5$ (owing to their intermittent and non-dispatchable nature) for the second crediting period and for subsequent crediting periods. Since project activity is of power generation by using Hydro energy, the above weightage has been considered for OM and BM.

Therefore,

$$EF_{grid,CM,y \text{ India}} = 0.9524 * 0.5 + 0.8687 * 0.5 \\ = 0.9105 \text{ tCO}_2/\text{MWh}$$

- Combined Margin Emission Factor for Nepal:

$$EF_{grid,CM,y \text{ Nepal}} = 0.0 \text{ tCO}_2/\text{MWh}^{33}$$

Nepal Electricity Generation³⁴	6052000	MWh
Power Purchase from India³⁵	2826000	MWh

- Connected system weighted average combined Emission Factor: =

$$\frac{\text{Nepal Generation} * EF_{grid,CM,y, \text{ Nepal}} + \text{Power purchase from India} * EF_{grid,CM, y, \text{India}}}{\text{Nepal Electricity Generation} + \text{Power Purchase from India}}$$

$$= 6052000 \times 0 + 2826000 \times 0.9105 / (6052000 + 2826000)$$

$$\text{Connected system weighted average combined Emission Factor} = 0.2898 \text{ tCO}_2/\text{MWh}.$$

5.2 Project Emissions

As per the approved consolidated Methodology AMS-I.D Version 18.0 para 39 “For most renewable energy project activities, $PE_y = 0$. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Where:

³³ <https://unfccc.int/documents/461676>

³⁴ https://www.nea.org.np/annual_report

³⁵ https://www.nea.org.np/annual_report

PE_y	= Project emissions in year y (t CO ₂ e/yr)
$PE_{FF,y}$	= Project emissions from fossil fuel consumption in year y (t CO ₂ /yr)
$PE_{GP,y}$	= Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (t CO ₂ e/yr)
$PE_{HP,y}$	= Project emissions from water reservoirs of hydro power plants in year y (t CO ₂ e/yr)

1. Samling Power Company Ltd. -

The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation $\left(PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}\right)$, is greater than 10 W/m²

$$\frac{9510000 \text{ W}}{18250 \text{ m}^2}$$

$$= 521.09 \text{ W/m}^2$$

As the calculated value from the given equation is 521.09 W/m² which is greater than 10 W/m², so as per applied methodology for the project activity, there would not be any project emission.

2. Asian Hydropower Ltd. -

The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation $\left(PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}\right)$, is greater than 10 W/m²

$$\frac{6200000 \text{ W}}{18250 \text{ m}^2}$$

$$= 339.72 \text{ W/m}^2$$

As the calculated value from the given equation is 339.72 W/m² which is greater than 10 W/m², so as per applied methodology for the project activity, there would not be any project emission.

5.3 Leakage

Since the project activity instances to be included into the grouped project activity are renewable energy projects based on hydropower, no leakage emissions are considered - in line with paragraph 27 of the small-scale methodology AMS.I.D (version 18.0).

5.4 Estimated Net GHG Emission Reductions and Removals

As per the para 28 of AMS-I.D version 18.0, the formula to calculate the emission reduction is

$$ER_y = BE_y - PE_y - LE_y$$

As the project activity is a hydropower project, there is no any leakage emissions from the project activity.

Hence, $LE_y = 0$

Therefore, Emission Reductions for this project activity are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y	=	Emission reductions in year y (t CO ₂ e/yr)
BE_y	=	Baseline emissions in year y (t CO ₂ e/yr)
PE_y	=	Project emissions in year y (t CO ₂ e/yr)

Therefore, Net GHG Emission Reductions and Removals are calculated as follows:

$$ER_y = BE_y - PE_y$$

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
<u>17-September-2021– 16-September-2022</u>	21,741	0	0	21,741
<u>17-September-2022– 16-September-2023</u>	21,741	0	0	21,741
<u>17-September-2023– 16-</u>	21,741	0	0	21,741

<u>September-2024</u>				
<u>17-September-2024-- 16-September-2025</u>	21,741	0	0	21,741
<u>17-September-2025-- 16-September-2026</u>	21,741	0	0	21,741
<u>17-September-2026-- 16-September-2027</u>	21,741	0	0	21,741
<u>17-September-2027-- 16-September-2028</u>	21,741	0	0	21,741
Total	152,187	0	0	152,187

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	EF_{grid, OM,y} (Indian Grid)
Data unit	tCO ₂ e/MWh
Description	Operating Margin CO ₂ emission factor in year y
Source of data	CO ₂ Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.

Value applied:	0.9524
Justification of choice of data or description of measurement methods and procedures applied	Calculated as the last 3-year (2019-20, 2020-21, 2021-22) generation-weighted average, sourced from Baseline CO ₂ Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.
Purpose of Data	For the calculation of the Baseline Emission
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	EF _{grid, BM,y} (Indian Grid)
Data unit	tCO ₂ e/MWh
Description	Build Margin CO ₂ emission factor in year y
Source of data	CO ₂ Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.
Value applied:	0.8687
Justification of choice of data or description of measurement methods and procedures applied	Calculated as the last year (2021-22) build margin, sourced from Baseline CO ₂ Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.
Purpose of Data	For the calculation of the Baseline Emission
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	EF _{grid, CM,y} (Indian Grid)
Data unit	tCO ₂ e/MWh
Description	Combined Margin CO ₂ emission factor in year y
Source of data	CO ₂ Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.

Value applied:	0.9105
Justification of choice of data or description of measurement methods and procedures applied	The combined margin emissions factor is calculated as follows: $EF_{grid,CM,y} = EF_{grid,OM,y} * W_{OM} + EF_{grid,BM,y} * W_{BM}$ Where: $EF_{grid,BM,y}$ = Build margin CO₂ emission factor in year y (tCO₂/MWh) $EF_{grid,OM,y}$ = Operating margin CO₂ emission factor in year y (tCO₂/MWh) W_{OM} = Weighting of operating margin emissions factor (%) = 50% W_{BM} = Weighting of build margin emissions factor (%) = 50%
Purpose of Data	For the calculation of the Baseline Emission
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	$EF_{grid,CM,y}$ (Nepal Grid)
Data unit	tCO _{2e} /MWh
Description	Combined Margin CO ₂ emission factor in year y
Source of data	Harmonized IFI Default Grid Factors 2021 v3.2 ³⁶
Value applied:	0.0
Justification of choice of data or description of measurement methods and procedures applied	NA
Purpose of Data	For the calculation of the Baseline Emission
Comments	This parameter is fixed ex-ante for the entire crediting period.

Emission factor of Nepal electricity system is not available as it is not published by Nepal Electricity Authority (NEA). Hence PP has taken the emission factor of Nepal is considered as zero from IFI

³⁶ <https://unfccc.int/documents/461676>

Default Grid Factors 2021 v3.2 published by UNFCCC³⁷. However, Nepal is importing electricity from India. Approximately 32% electricity is imported from Indian grid and emission factor of Indian grid is calculated as 0.9105 by using CEA database³⁸ as per TOOL07. Hence the emission factor of Nepal is calculated as weighted average emission factor of Nepal and India. Asian Development Bank has published the emission factor of Nepal is 348 tCO₂/ GWh³⁹. Weighted average of Nepal and India grid is taken as 0.2898 tCO₂/ MWh (i.e. 289 tCO₂/ GWh) which is lower than Asian Development Bank and should be considered as a conservative.

6.2 Data and Parameters Monitored

Data / Parameter	EGPJ, y
Data unit	MWh/y
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Credit note/ JMR/B Form reports/ monthly generation report from state electricity board/DISCOM
Description of measurement methods and procedures applied	-
Frequency of monitoring/recording	Continuous measurement & monthly recording
Value applied:	75,016 (Estimated value)
Monitoring equipment	Type: Electricity Meter Accuracy of meters: 0.1s
QA/QC procedures applied	Meters will be calibrated once in 5 years
Purpose of data	Calculation of baseline emissions
Calculation method	The Joint Meter Readings (JMR) taken every month from each meter are added up to arrive at the net value of electricity generated by project activity. Also bills of net metering will be

³⁷ <https://unfccc.int/documents/461676>

³⁸ https://cea.nic.in/wp-content/uploads/baseline/2023/01/Approved_report_emission_2021_22.pdf

³⁹ <https://www.adb.org/sites/default/files/linked-documents/49086-001-crf.pdf>

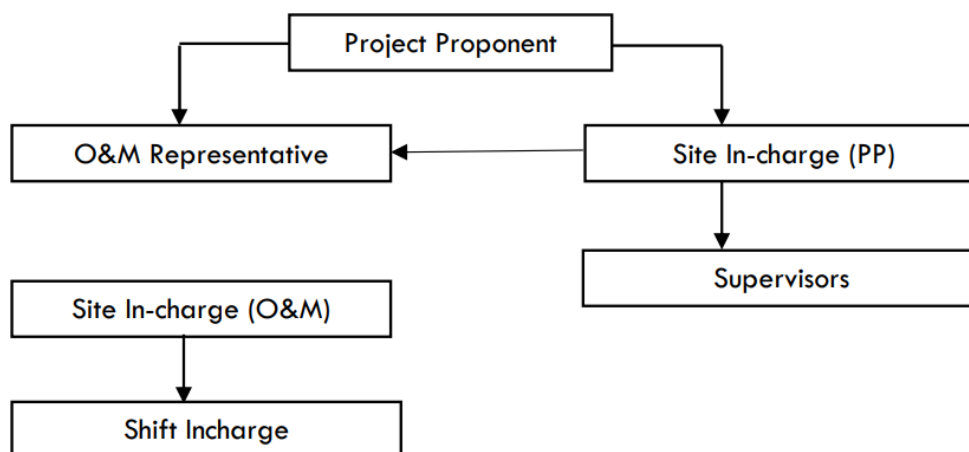
	considered for determination of net electricity generated by project activity.
Comments	-

6.3 Monitoring Plan

The monitoring plan is developed in accordance with the modalities and procedures for CDM project activities. The monitoring plan, which will be implemented by the project participant describes about the monitoring organisation, parameters to be monitored, monitoring practices, quality assurance, quality control procedures, data storage and archiving.

Organizational structure and responsibilities:

The authority and responsibility for registration, monitoring, measurement, reporting and reviewing of the data rests with the project participant. PP proposed the following structure for data monitoring, collection, data archiving and calibration of equipment for this project activity. The team comprises of the following members:



Responsibilities of Site Incharge (PP): Overall functioning and maintenance of the project activity, the Site incharge shall coordinate with the O&M operator as well as the site supervisors.

Responsibilities of O&M Representative: Co-ordination between Site incharge of the O&M operator as well as the project participant and further report to PP head office.

Responsibilities of Site In-charge (O&M Operator): Responsibility for maintaining the data records, ensures completeness of data, and reliability of data (calibration of equipment) as well as data recording for all the parameters.

Responsibilities of Shift In-charge: Responsibility for day-to-day data collection and maintains day to day monitored data.

Data Measurement

The export and import energy or net electricity displaced by project will be measured continuously using above mentioned energy meter located after inverters. Readings of meters shall be taken on monthly basis by authorized officer of developer in the presence of PP or representative of PP. In the event of failure of generation meter, the user energy meter or inverter generation value will be used in monitoring the electricity generation data.

Data collection and archiving

Readings from meters will be collected in the presence of the plant in-charge. Export and import or net electricity displaced by project would be recorded and stored in logs as well as in electronic form on a daily basis. The records are checked periodically by the Plant Manager and discussed thoroughly with the plant supervisor. The period of storage of the monitored data will be 2 years after the end of crediting period or till the last issuance of VERs for the project activity whichever occurs later.

Emergency preparedness

The project activity will not result in any unidentified activity that can result in substantial emissions from the project activity. No need for emergency preparedness in data monitoring is visualized.

QA/QC Procedures

There energy meters are installed at the respective project site. The calibration of all the meters will be undertaken at required intervals (once is five years as per CEA notification) and faulty meters will be duly replaced immediately. Records of calibration certificates will be maintained for verification purposes. Hence, a reliable method will be ensured with monitoring of the parameters. The invoice records will be used and kept for cross checking the consistency of the recorded data.

Sampling Approaches

Not required

Personnel training

In order to ensure a proper functioning of the project activity and a properly monitoring of emission reductions, the staff will be trained. The plant helpers will be trained in equipment operation, data recording, reports writing, operation and maintenance and emergency procedures in compliance with the monitoring plan.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Value applied:	91,678.73
Comments	-

7.2 Baseline Emissions

Baseline emission is the product of Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity multiplied by the Combined margin CO₂ emission factor for grid connected power generation

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where,

BE_y = Baseline Emissions in year y; t CO₂

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system Version 7.0” (t CO₂/MWh).

7.3 Project Emissions

As per methodology AMS I.D Version 18.0, for renewable energy projects, there is no any project emissions occurred.

Hence, $PE_y = 0$

7.4 Leakage

As per methodology AMS I.D Version 18.0, for renewable energy projects, there is no any leakage emissions occurred.

Hence, $LE_y = 0$

7.5 Net GHG Emission Reductions and Removals

The total emission reduction for current monitoring period (17-September-2021 to 31-December-2022) is = 26,567 tCO_{2e}.

The total monitoring period (17-September-2021 to 31-December-2022) is of 365 days. The estimated annual emission reduction as per PD is 21,741 tCO₂ for first crediting period.

So, the estimated emission reduction for the 471 days of monitoring period will be

$$= (21,741/365) \times 471$$

$$= 28,055 \text{ tCO}_{2e}$$

The % fall in emission reduction in current monitoring period as compared to estimated emission reduction is = $(26567 - 28055) / 28055$

$$= - 0.0530$$

$$= - 5.30\%$$

The net GHG emission reduction for this project activity for current monitoring period is shown table below:

Table: Net Emission Reduction from 17-September-2021 to 31-December-2022

Year	Baseline emissions or removals (tCO _{2e})	Project emissions or removals (tCO _{2e})	Leakage emissions (tCO _{2e})	Net GHG emission reductions or removals (tCO _{2e})
17-September-2021 to 31-December-2021	2,239	0	0	2,239
01-January-2022 to 31-	24,328	0	0	24,328

December -2022				
Total	26,567	0	0	26,567

<u>Ex-ante emissions reductions /removals</u>	<u>Achieved emissions reductions /removals</u>	<u>Percent difference</u>	<u>Justification for the difference</u>
28,055	26,567	-5.30%	The actual emission reduction is -5.30% lesser than the estimated value which is due to variation of water availability in Canal (where the project activity is installed). The water availability in canal is get affected by rainfall, trash or monsoon season, hence, this factor is beyond the control of the PP.